21-242 Recitation 12

Dimension — Q1. An independent list is no longer than a basis

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that the length of a linearly independent list is at most the length of a basis.

Course-level solution

Let $(\vec{v}_1, \ldots, \vec{v}_k)$ be linearly independent in V, and let $(\vec{w}_1, \ldots, \vec{w}_n)$ be a basis of V. We show $k \leq n$ by the replacement (Steinitz) argument.

The list $(\vec{v}_1, \vec{w}_1, \ldots, \vec{w}_n)$ is spanning (it still spans V) and linearly dependent ($\vec{v}_1$ is in the span of the $\vec{w}$'s). In a dependent spanning list, some vector is in the span of the preceding ones; that vector cannot be $\vec{v}_1$ (independence of a single nonzero vector, or $\vec{v}_1 \neq \vec{0}$), so some $\vec{w}_{j_1}$ is in the span of $\{\vec{v}_1\} \cup \{\vec{w}_i : i \neq j_1\}$. Drop it: $(\vec{v}_1, \vec{w}_1, \ldots, \widehat{\vec{w}_{j_1}}, \ldots, \vec{w}_n)$ still spans.

Repeat: adjoin $\vec{v}_2$ on the left. The list is dependent, and sequential independence of $(\vec{v}_1, \vec{v}_2)$ means the vector we drop is one of the remaining $\vec{w}$'s. After k steps we have replaced k of the $\vec{w}$'s by $\vec{v}_1, \ldots, \vec{v}_k$ and still have a spanning list. We cannot have run out of $\vec{w}$'s before using all $\vec{v}$'s: otherwise $(\vec{v}_1, \ldots, \vec{v}_k)$ would be dependent. Hence $k \leq n$.

Later-course solution (using ideas from the rest of the handout)

After linear maps: independence of $(\vec{v}_1, \ldots, \vec{v}_k)$ means the map $\mathbb{F}^k \to V$, $e_i \mapsto \vec{v}_i$ is injective, so $k = \dim \mathbb{F}^k \leq \dim V = n$ once one knows injective maps cannot raise dimension (Dimension Q3 / rank-nullity). That proof is shorter but logically later.